

Childhood Cancer getting better, getting trickier

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Health Claims Forum

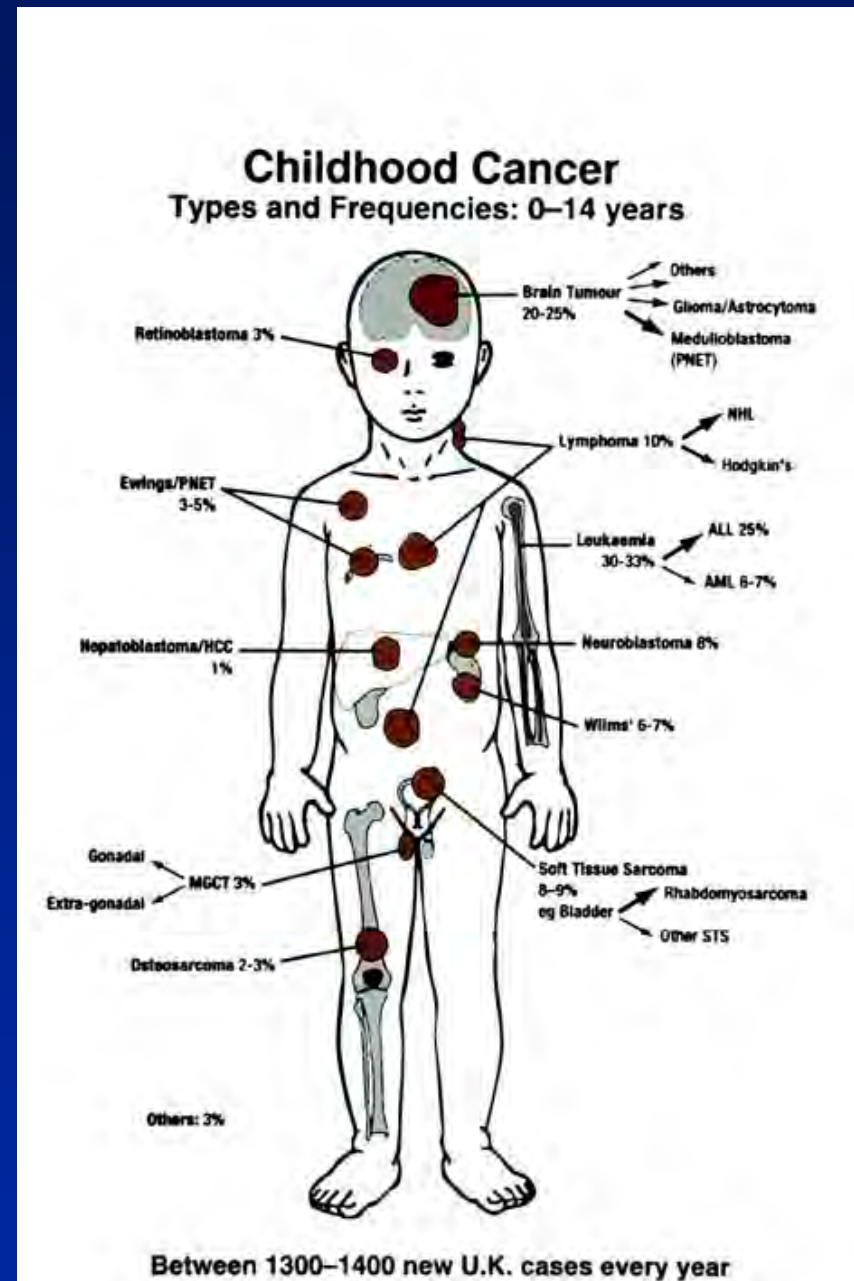
16.02.2017

Childhood cancer - mercifully rare:

- **1 in 600 children develop cancer by 15 years of age**
- **1,200 new cases in the UK per annum**
- **0.5% of cancer cases occur in childhood**
- **but second commonest cause of death in childhood**

Childhood cancer-
different to adults:

different sites
different histology
different age ranges



The bad old days....

- Universally fatal
- Poorly recognised
- Symptoms difficult to manage and control

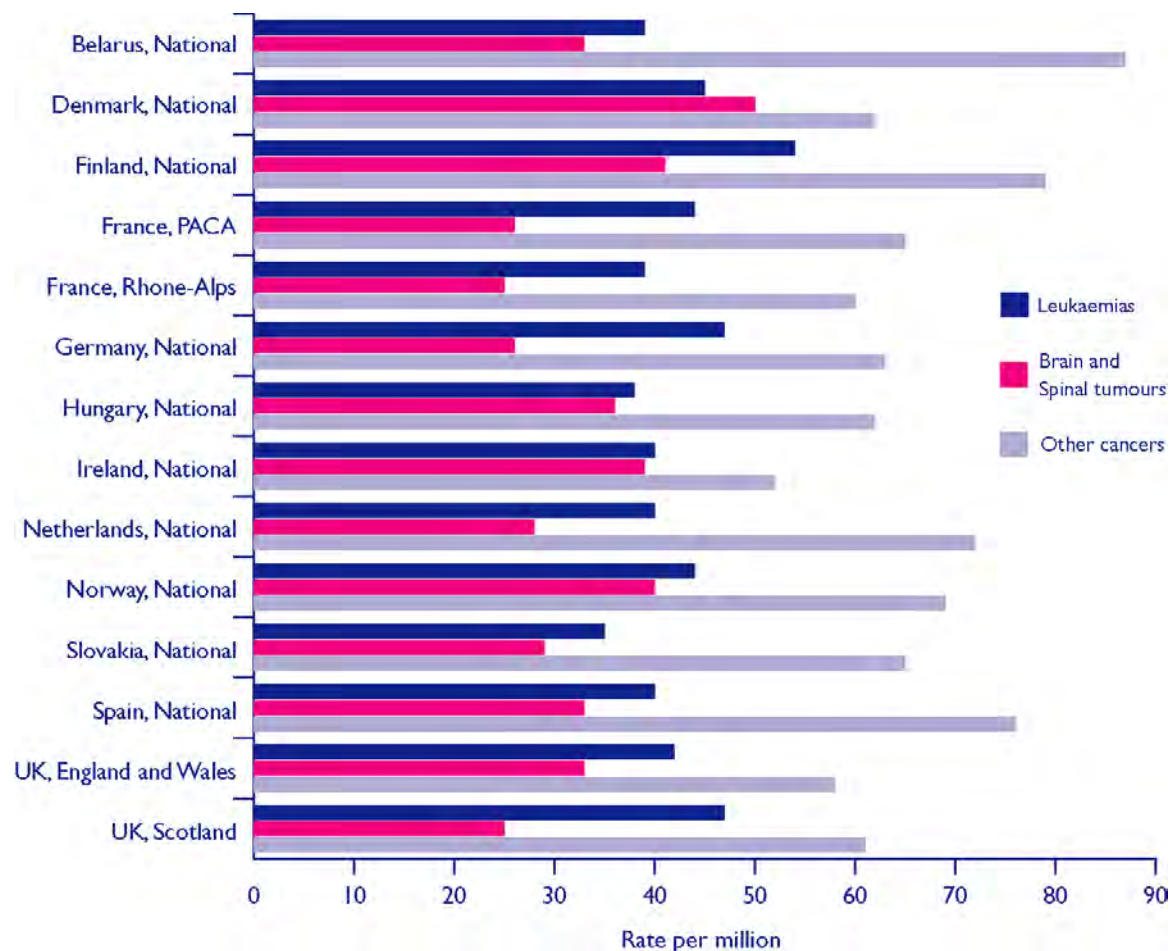


Structure of presentation

- What sort of cancer do children get?
- Incidence by age and cancer type – congenital tumours
- What is cancer?
- Is our terminology right?
- How do we assess risk?
- Novel therapies
- Late effects of therapy
- Is there a risk to family members?

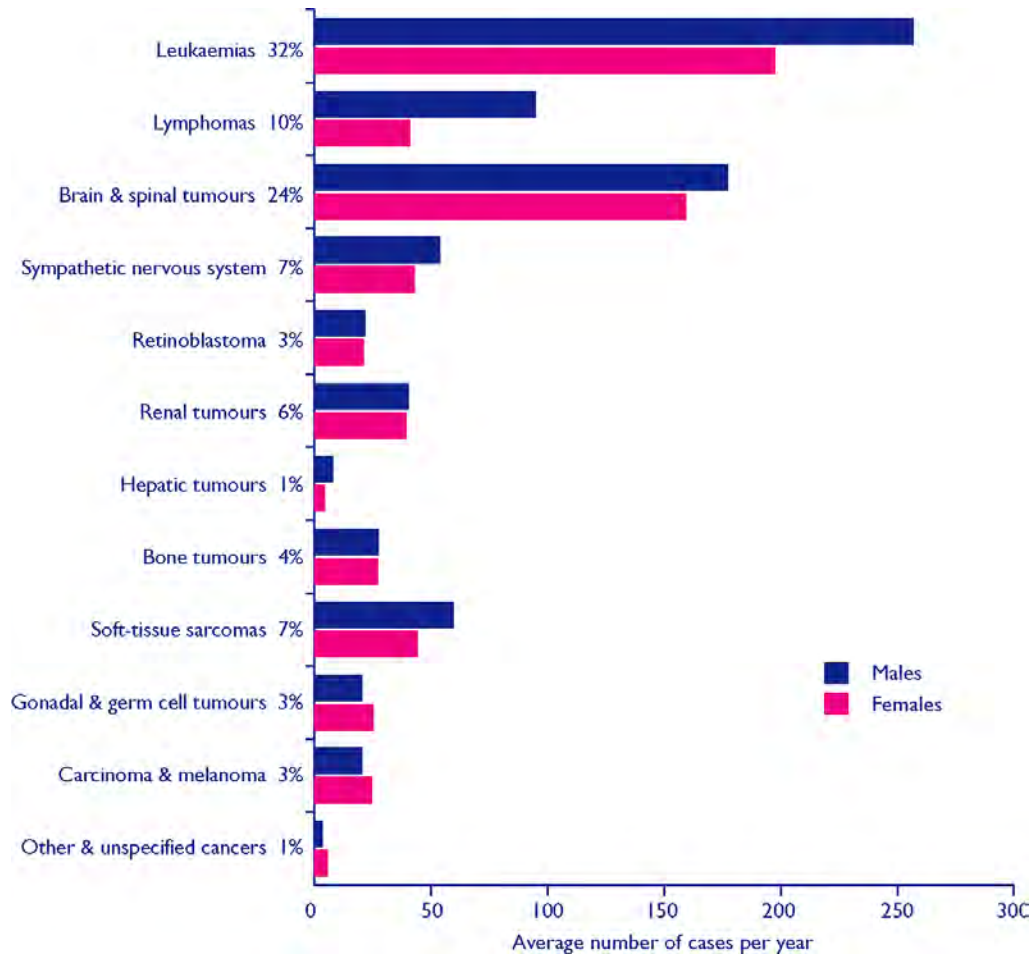
**Are there differences in incidence
internationally?**

European age standardised incidence rates for cancers diagnosed at ages 0-14



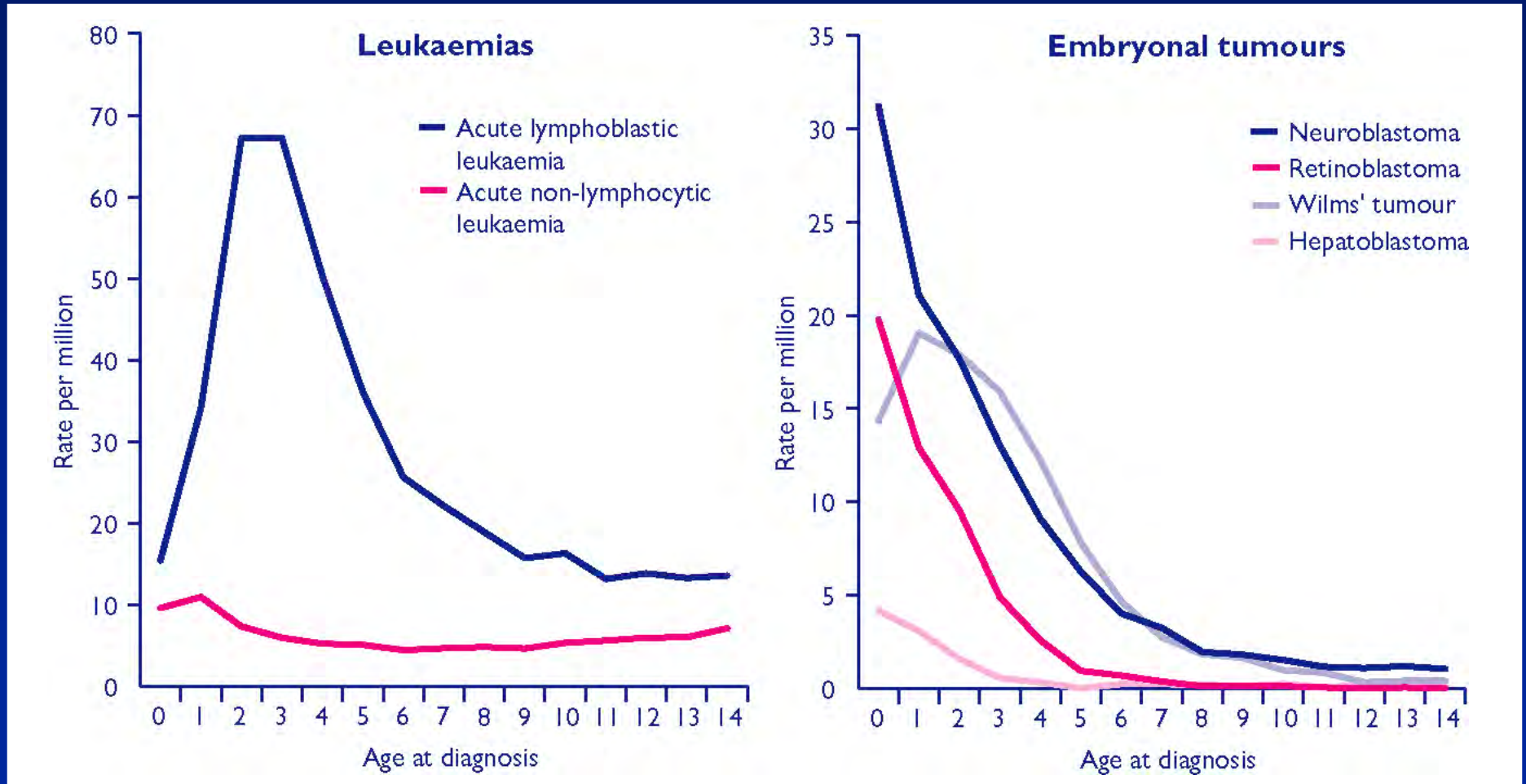
What types of cancer do children get?

Annual average number of cases by diagnostic group and sex, ages 0-14

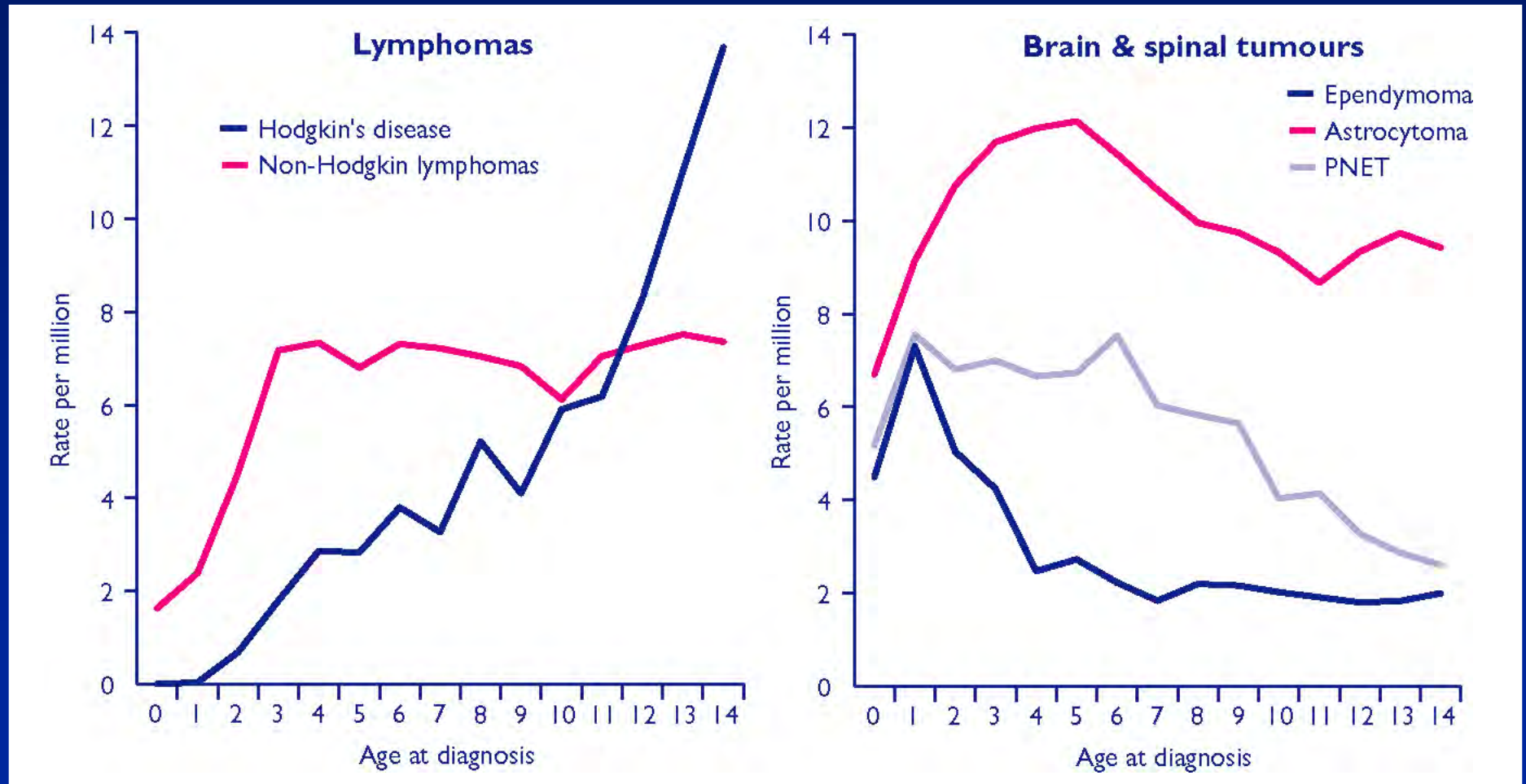


When do cancers occur in childhood?

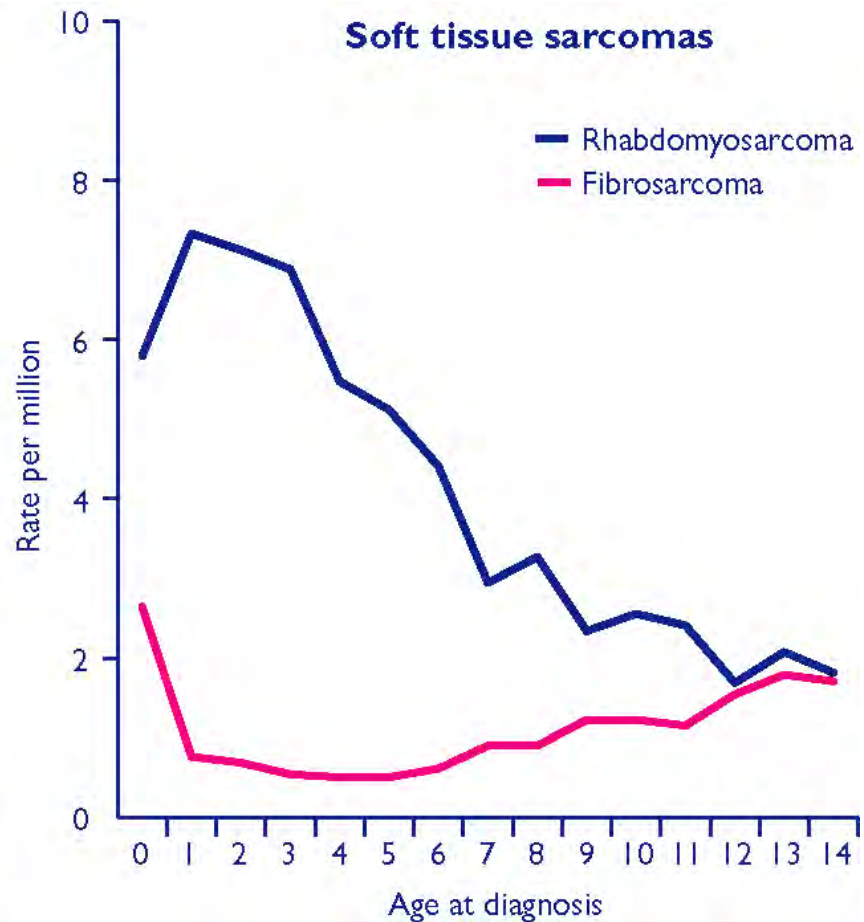
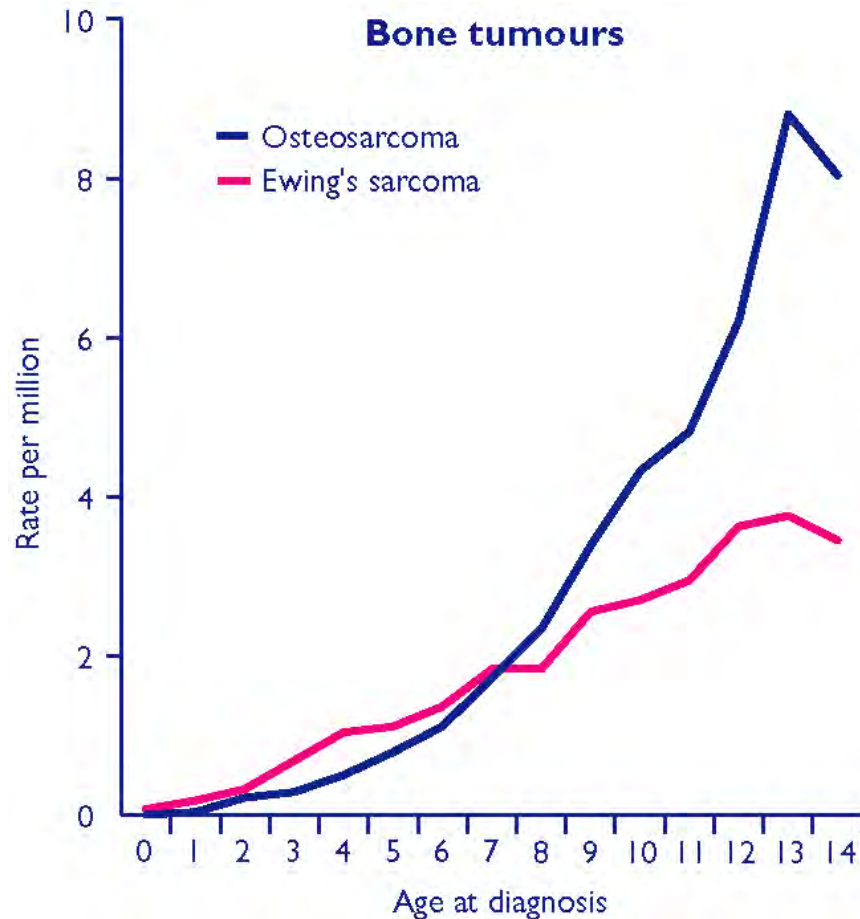
Age-specific incidence rates per million, selected childhood cancers, Great Britain 1962-97



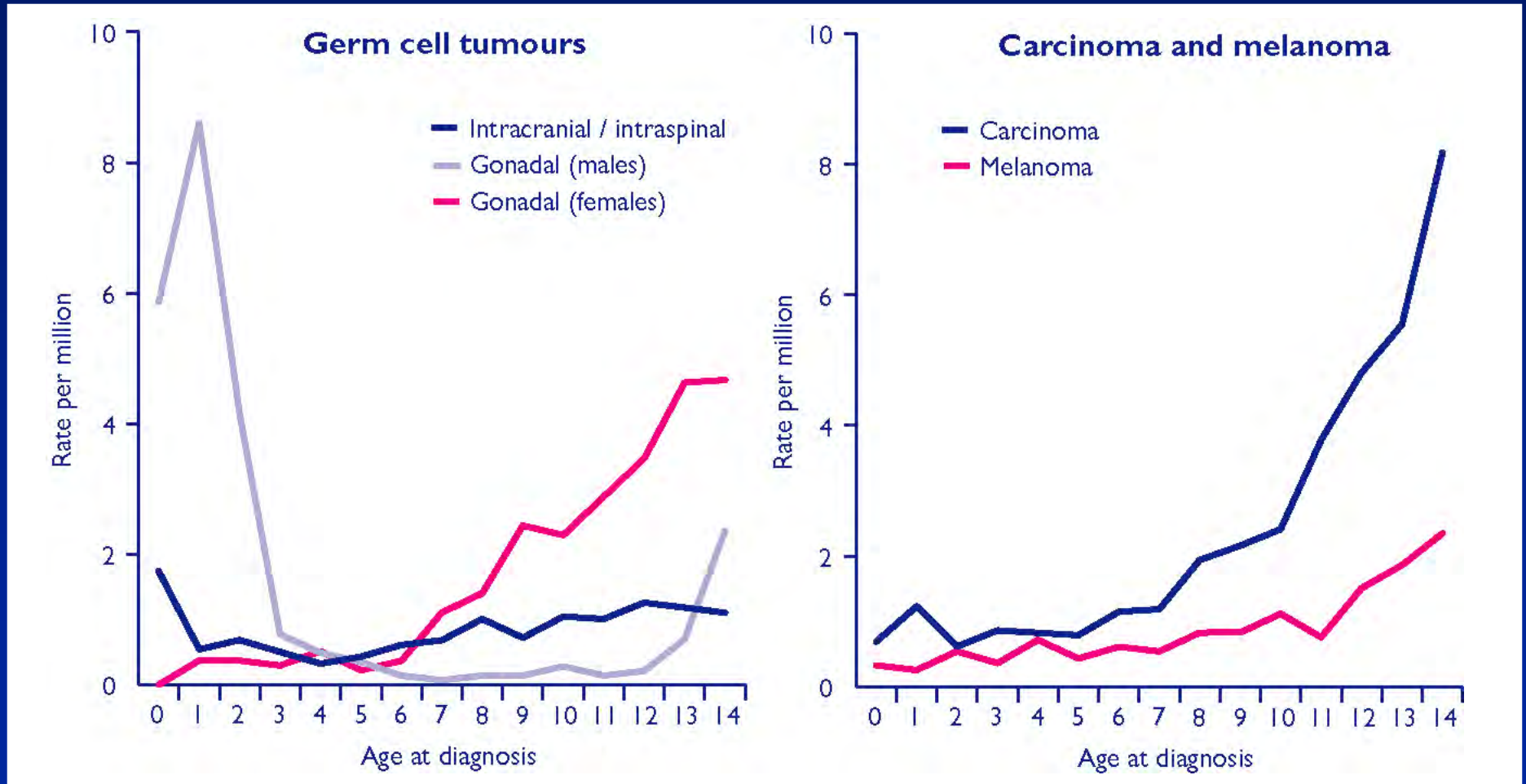
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Age-specific incidence rates per million, selected childhood cancers, Great Britain 1962-97



Congenital or not?

Definitely Congenital

- **Antenatal diagnosis of a mass:**
 - Neuroblastoma – often just observed
 - Brain tumour leading to difficulty in delivery – often benign tumours
- **Obvious at delivery**
 - Teratoma – often sacrococcygeal – surgery only for most

Congenital....but

- **Downs syndrome – TAM**
(transient abnormal myelopoiesis) arises day 3 –
can settle spontaneously or with gentle
chemotherapy
- **Infantile sarcomas – mass arises within first few**
days/weeks of birth but not obvious at birth

Not congenital ...but

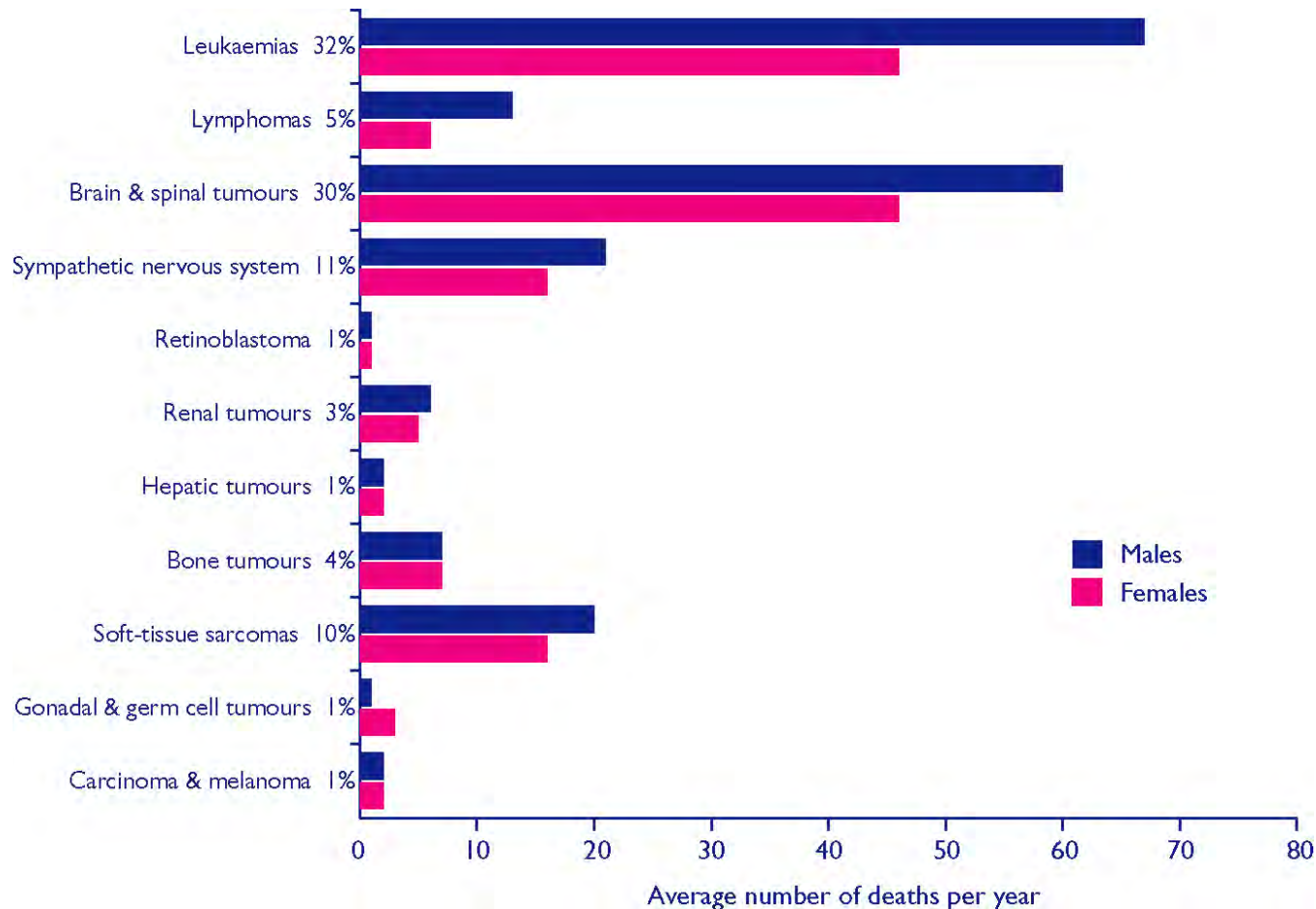
- **Acute myeloid leukaemia – abnormal translocation found on Guthrie blood spots in children who only go on to develop disease several years later**
- **Neuroblastoma – screening by urine samples detects children who would never go on to develop clinical disease**

What is impact of cancer on childhood mortality?

Main causes of child mortality, ages 1-14, by sex and age group, England and Wales, 2000-2002

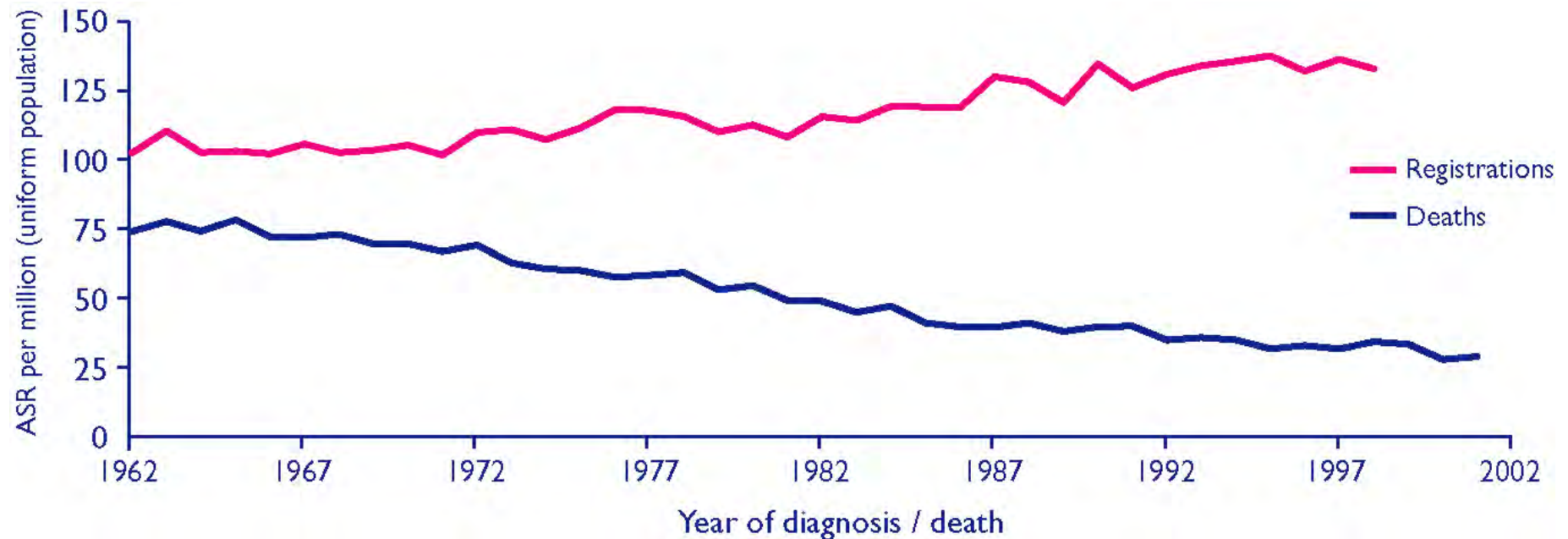
Cause of death	Males		Females	
	% of total	% of total	% of total	% of total
	1-4 years	5-14 years	1-4 years	5-14 years
Infections	11	3	7	5
Cancers	14	23	13	24
Nervous system and sense organs	14	15	12	14
Circulatory system	4	5	7	6
Respiratory system	9	5	8	8
Congenital anomalies	14	7	15	9
Accidents	17	30	16	19
Other	17	12	20	16
Total number of deaths	968	1,444	752	1,049

Annual average number of deaths in children aged under 15 diagnosed with cancer, by diagnostic group and sex.

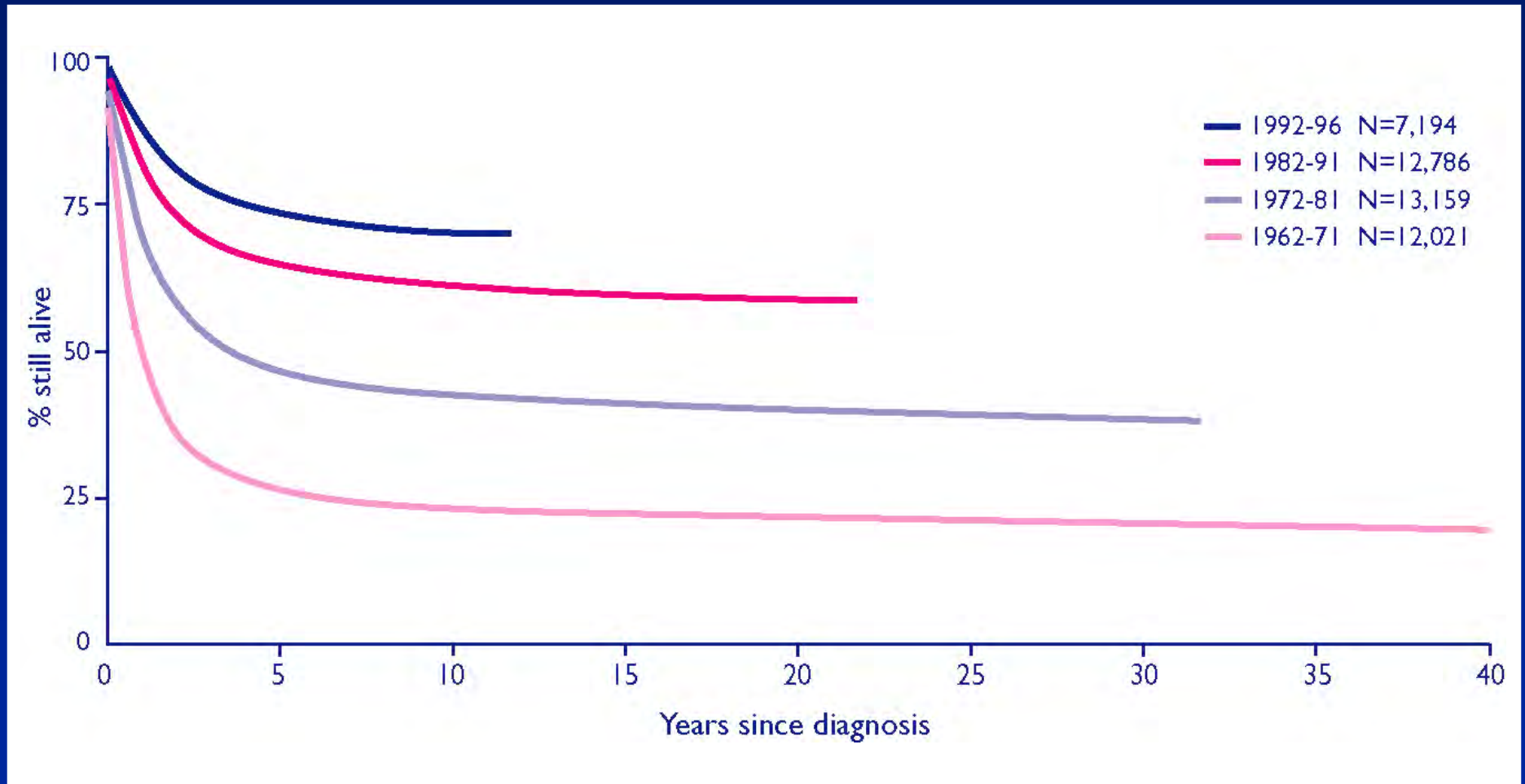


Are results improving?

Trends in age standardised (uniform population) cancer registration and death rates, children and 0-14 years, Great Britain, 1962-1998/2001



Survival of childhood cancer patients diagnosed in successive periods, Great Britain, 1962-96



**What do we understand about
childhood cancer?**

Cancer



Normal Cell

**GROW WHEN TOLD
ALWAYS CHECK**

GROW

Gene Amplification

**GROW GROW GROW
GROW WHEN TOLD
ALWAYS CHECK**

GROW GROW

Mutated Message

GROW WHEN TOLD
ALWAYS CHECK

GROW

Increased Expression

**ALWAYS GROW
CHECK WHEN TOLD**

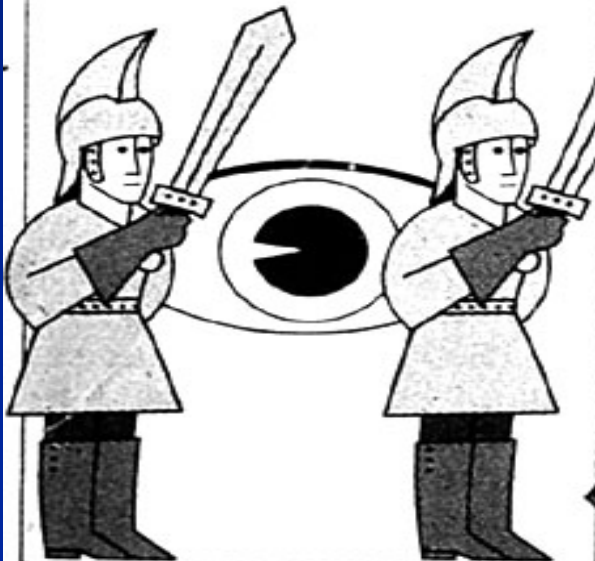
GROW

GROW

Two hit hypothesis

HOW CANCER ATTACKS THE EYE

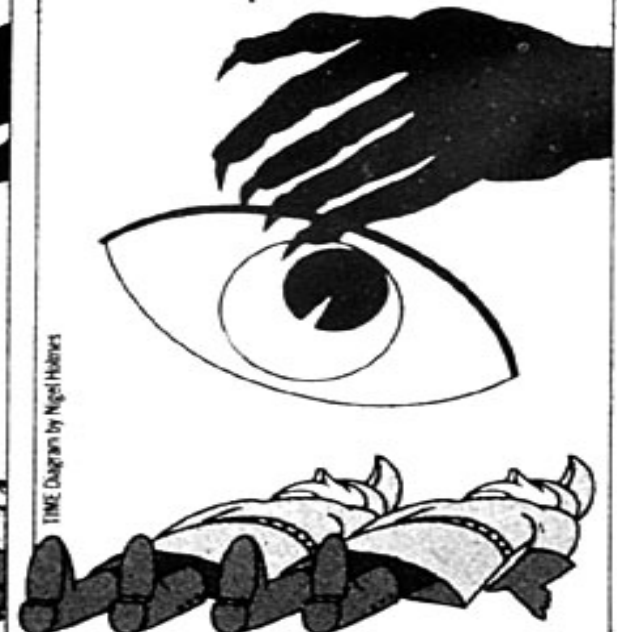
1 Most people are born with two anticancer genes that act like guardians to protect the eye from cancer.



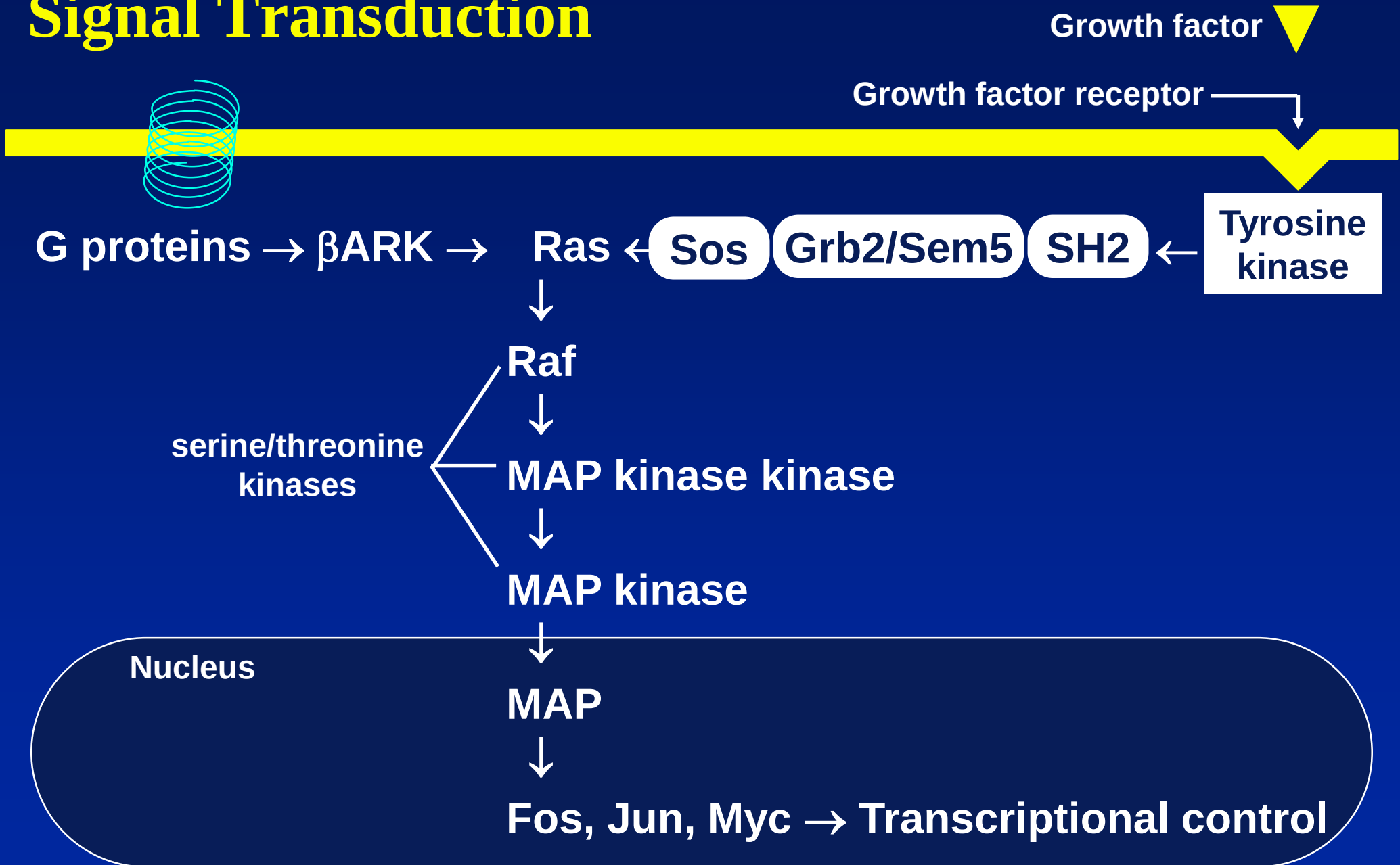
2 Through heredity, one gene may be defective or missing. The other is only just strong enough to prevent cancer from attacking...



3 ... but if some biological accident disables this gene, cancer could develop.



Signal Transduction



Cancer



Why is molecular biology useful?

- Makes a diagnosis
- Allows identification of a risk group
- Molecular abnormalities are targets for new therapies
- Defines genetic risk to family members
- All of oncology will be based on an understanding of molecular biology – it is not optional!

Making a diagnosis

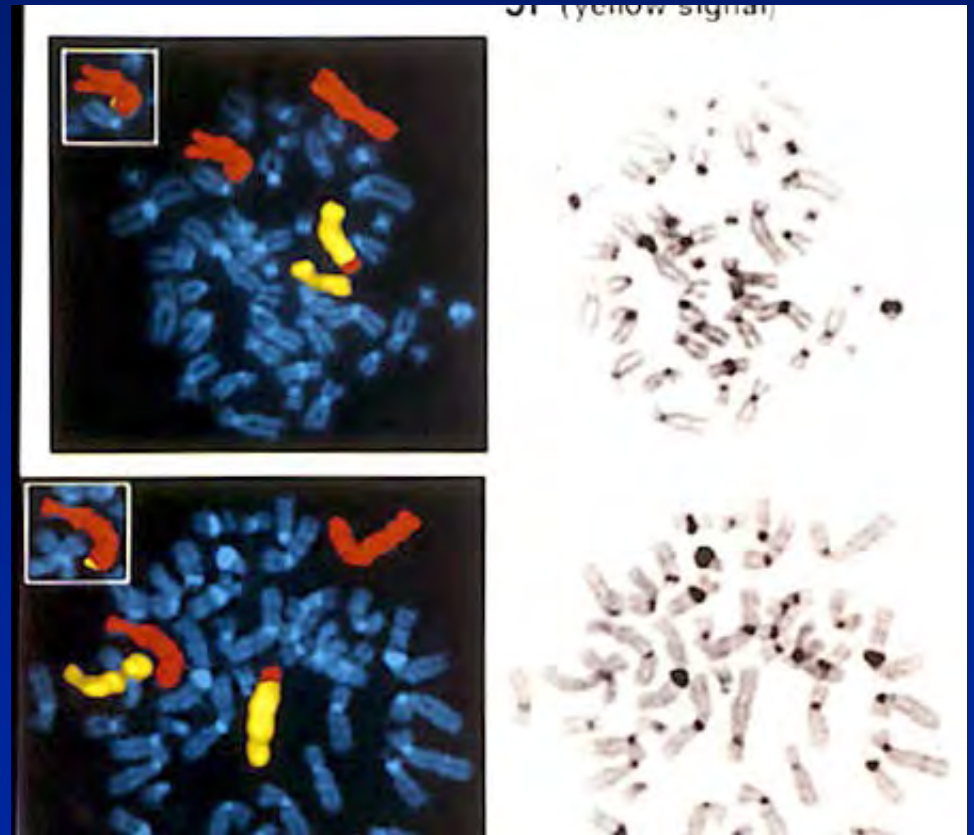
- **Histology is sometimes difficult to interpret**
- **Some tumours have characteristic chromosomal translocations that can be identified by FISH or PCR**
 - Alveolar rhabdomyosarcoma t(2;13)
 - Ewings/pPNET t(11;22)
- **Gene expression profiling gives 'origin' of tumour**
 - 450k array or affymetrix panels
- **Direct sequencing can identify mutations**
 - retinoblastoma, BRAF, NF1

Diagnostic molecular testing

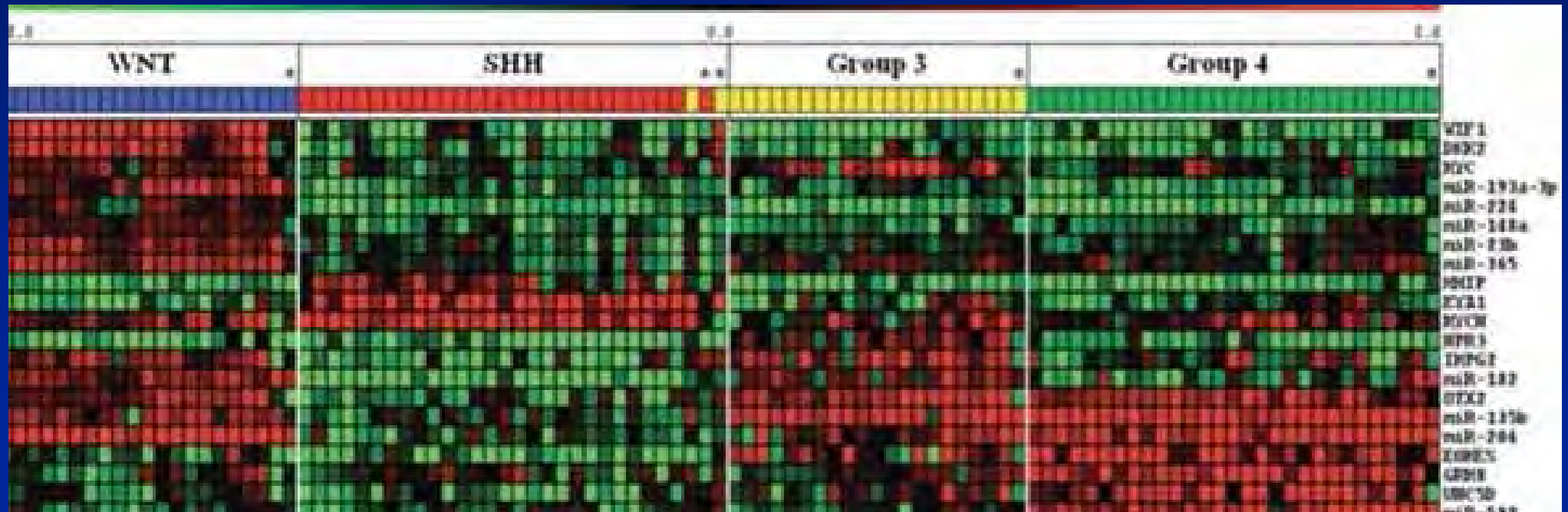
Fish



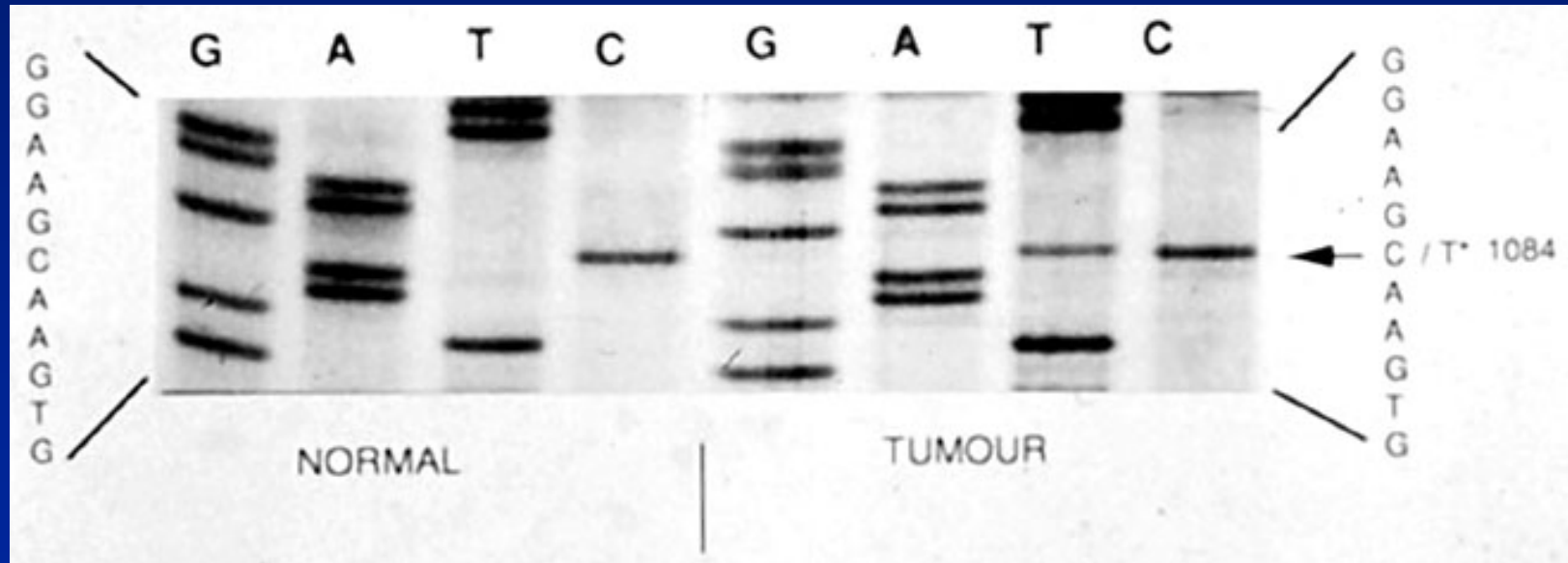
FISH



Diagnostic molecular testing- gene expression profiling medulloblastoma



Diagnostic testing sequencing gene in retinoblastoma



Diagnostic categories have changed...

- Cancer or not?
- Benign v malignant
- TNM stage, Tumour grade

Which is a cancer?

Hodgkins LP

- Grade 4 WHO

LCH

- Not classified as a tumour at all in WHO

Which is a cancer?

Hodgkins LP

- Grade 4 WHO
- Surgery only
- Good chance of cure

LCH

- Not classified as a tumour at all in WHO
- Can have multisystem involvement
- Most need chemotherapy
- Some need high dose chemotherapy with stem cell rescue or novel targeted agents

Which is 'benign'?

Stage 4S neuroblastoma

- Grade 4 tumour WHO
- Metastatic to liver and skin
- Rapidly growing

Craniopharyngioma

- Grade 1 (Benign) WHO
- Never spreads
- Grows slowly

Which is 'benign'?

Stage 4S neuroblastoma

- Grade 4 tumour WHO
- Metastatic to liver and skin
- Rapidly growing
- Resolves spontaneously with usually no/minimal late sequelae

Craniopharyngioma

- Grade 1 (Benign) WHO
- Never spreads
- Grows slowly
- Surgery, radiation therapy. Blindness and a life time of endocrine problems

Terminology



- TNM not useful or used in paediatrics
- WHO grade irrelevant and not prognostic
- Risk stratification based on different criteria
- How are you going to assess claims?

Genetics can help to give a risk group

- How do we find out what is relevant?
- Traditionally:
 - What it has done already
 - What it looks like

What will they do today?



Using biology to predict risk



European Clinical trials & Biological studies

Trial

PNET3

1992-2000

PNET4

2000-2007

PNET5

2013-2019

Biological study

**Marker Identification
(FFPE)**

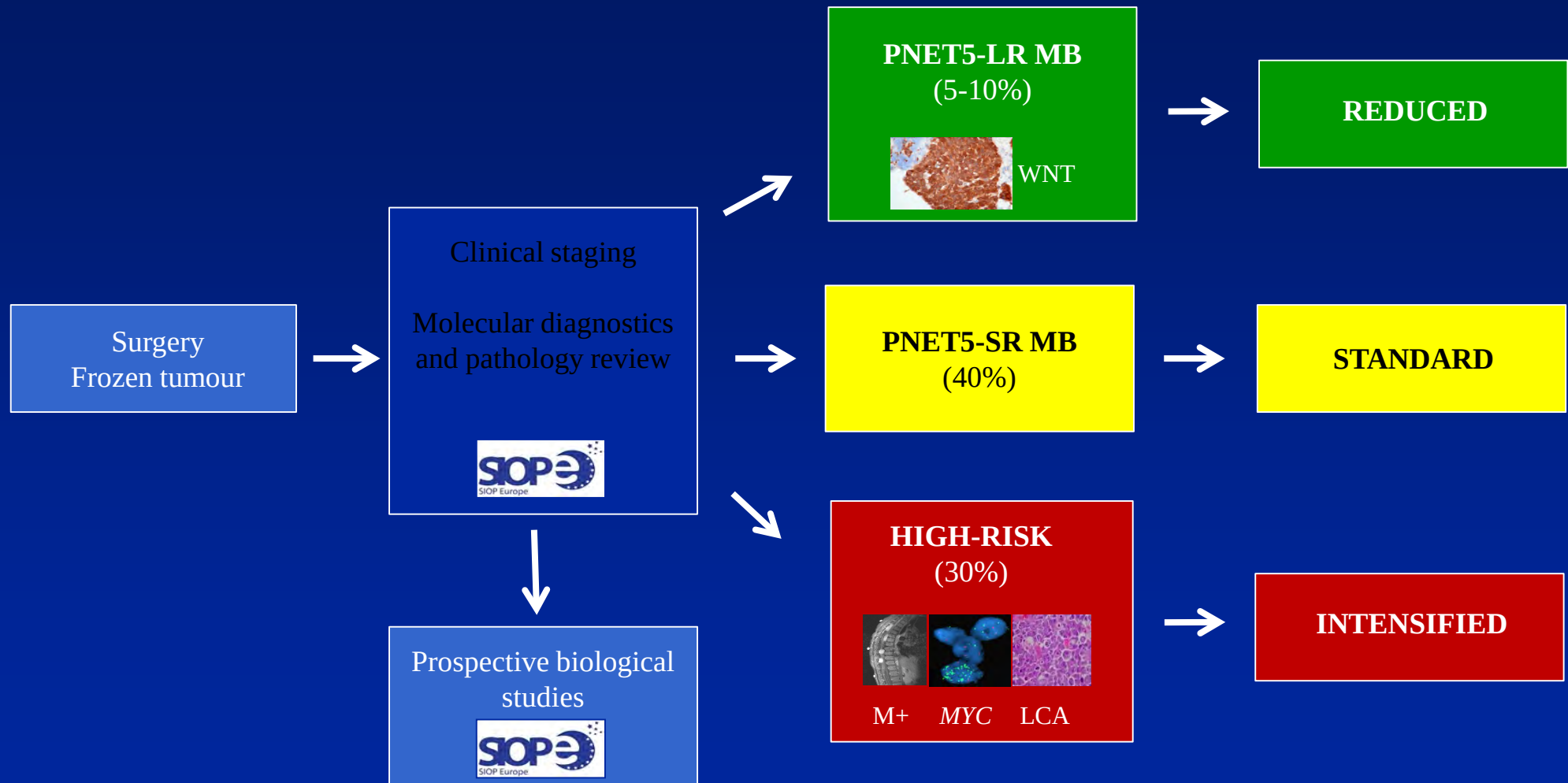
**Marker validation
(FFPE)**

Marker selection

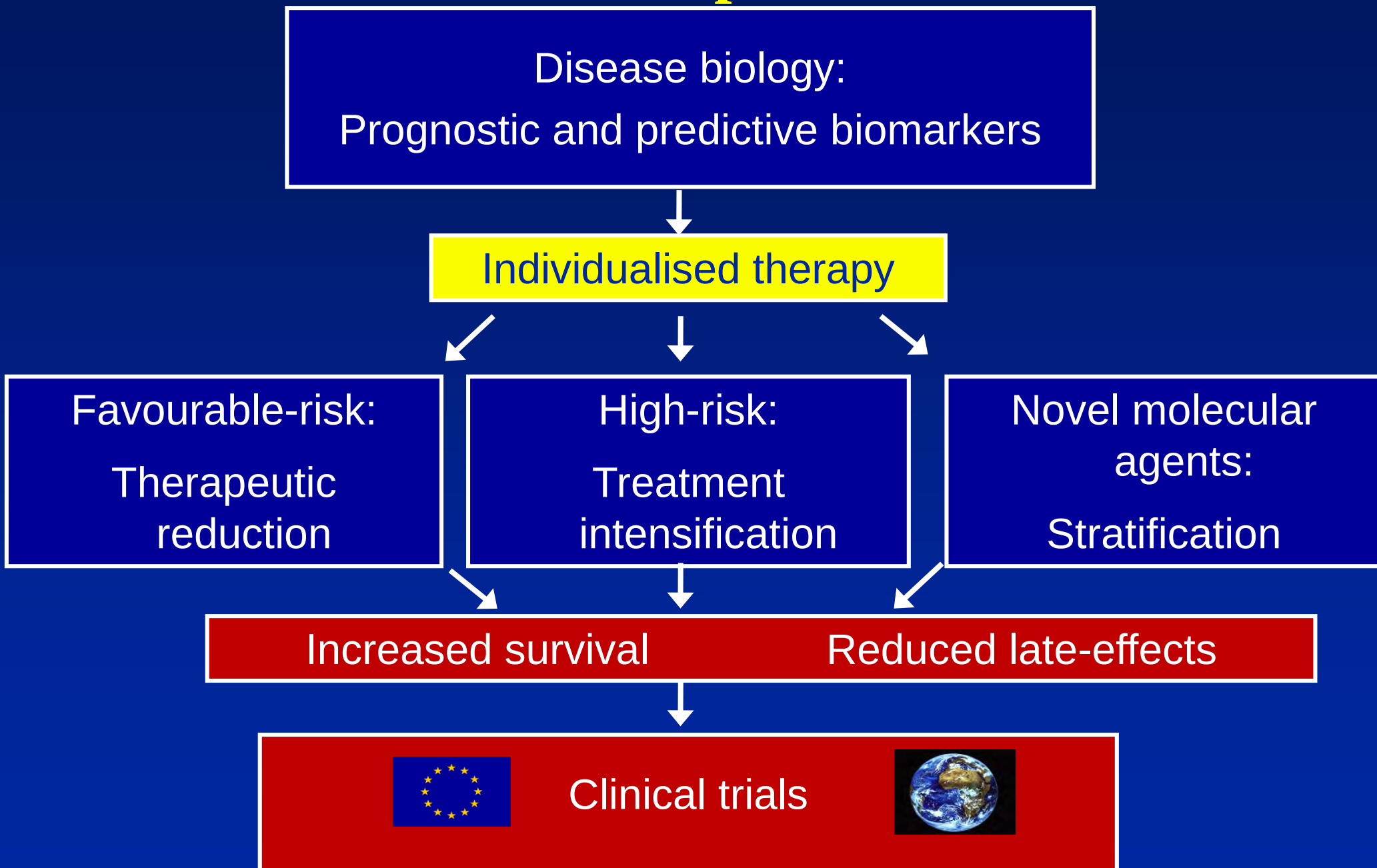
**New markers & drug
targets (Frozen)**



The PNET5 MB clinical trial (2013-2019)



Biomarker-driven therapies for CNS Tumours



Medulloblastoma subgroups: Current consensus

Molecular Subgroups of Medulloblastoma				
CONSENSUS	WNT	SHH	Group 3	Group 4
Cho (2010)	C6	C3	C1/C5	C2/C4
Northcott (2010)	WNT	SHH	Group C	Group D
Kool (2008)	A	B	E	C/D
Thompson (2006)	B	C ¹ , D	E, A	A, C
DEMOGRAPHICS				
Age Group:   	  	    	  	    
Gender: ♀ ♂	♂ ♂ : ♀ ♀	♂ ♂ : ♀ ♀	♂ ♂ : ♀	♂ ♂ : ♀
CLINICAL FEATURES				
Histology	classic, rarely LCA	desmoplastic/nodular, classic, LCA	classic, LCA	classic, LCA
Metastasis	rarely M+	uncommonly M+	very frequently M+	frequently M+
Prognosis	very good	infants good, others intermediate	poor	intermediate
GENETICS				
	 CTNNB1 mutation	 PTCH1/SMO/SUFU mutation GLI2 amplification MYCN amplification	 i17q MYC amplification	 i17q CDK6 amplification MYCN amplification
GENE EXPRESSION				
	WNT signaling MYC +	SHH signaling MYCN +	Photoreceptor/GABAergic MYC +++	Neuronal/Glutamatergic minimal MYC / MYCN

Terminology

Traditional

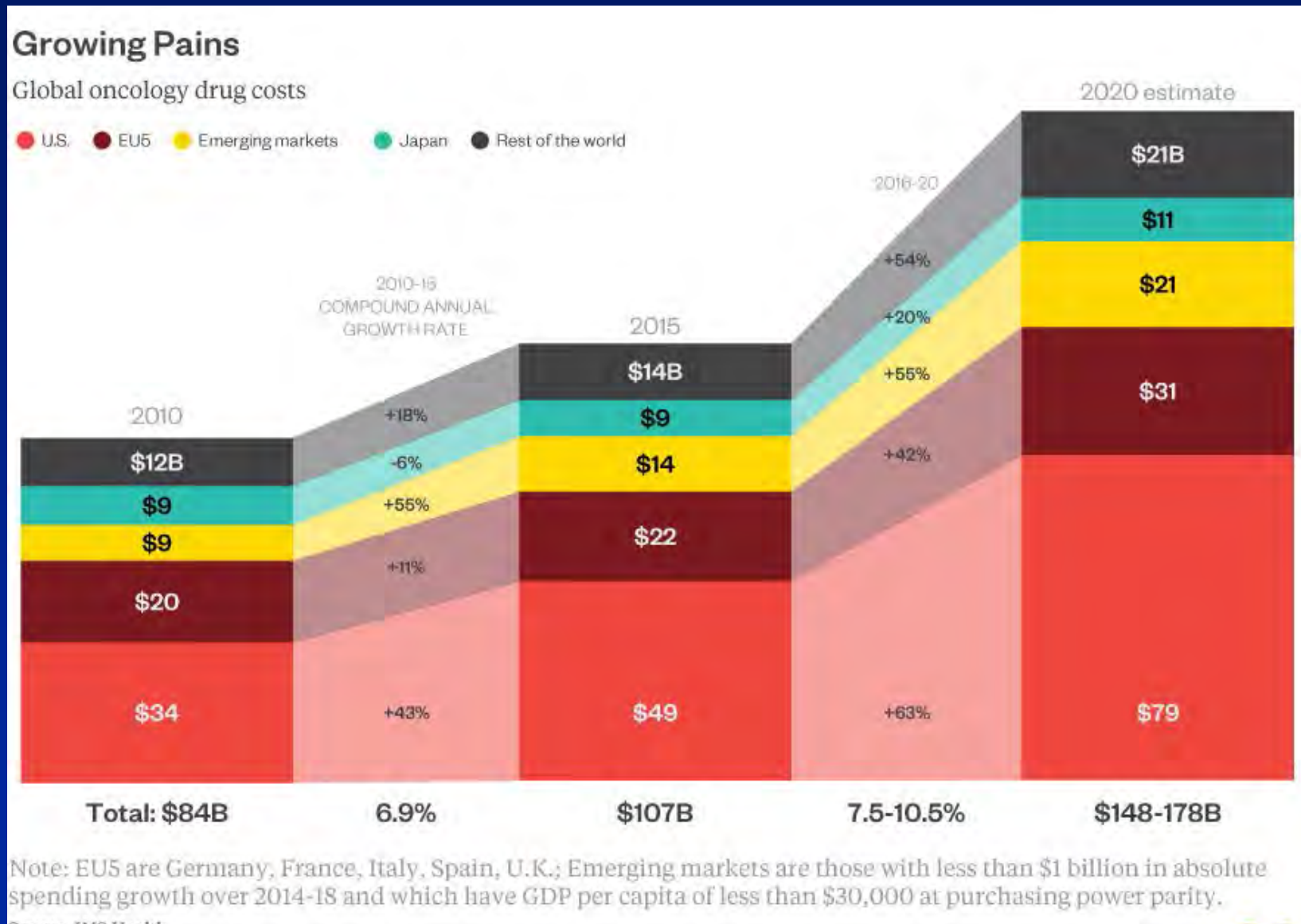
- Medulloblastoma
- Group 4 WHO
- No spread

Now

- Patient 1:
 - Large cell group 3 = 5% EFS
- Patient 2:
 - Desmoplastic group 2 = 90% EFS

**Genetics can identify new targets for
therapy**

Cost of new approaches...



**Blocking upstream doesn't always work-
multiple agents may be necessary**

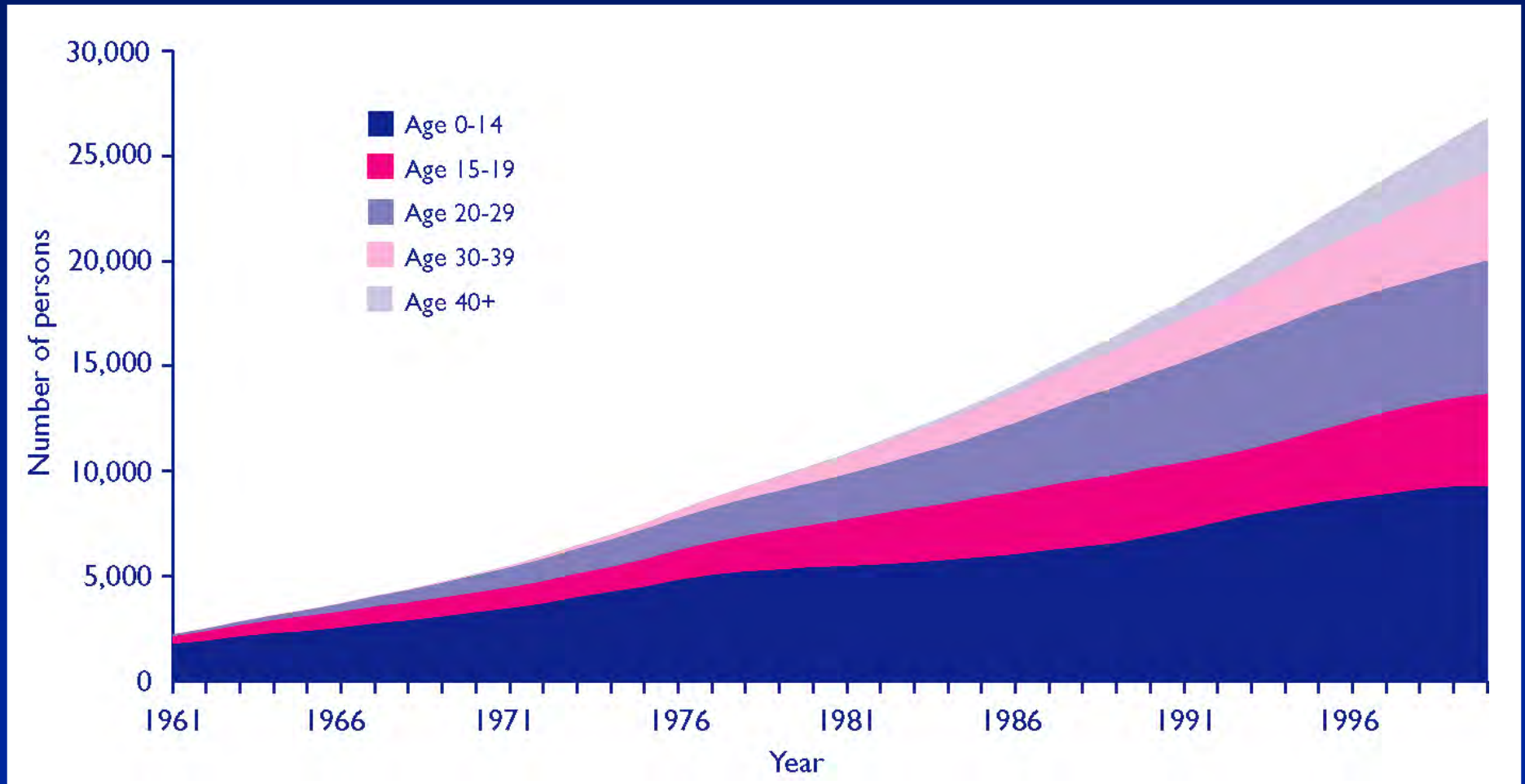


Was it in their genes?

- Traditionally <5% of paediatric cancer thought to be due to germline mutations
- Now >8% of patients carry germline mutations of currently known predisposition genes
- Multi-gene profiles will predict risk with ever more accuracy
- Impact on individual (risk in future)
- Impact on family (are they insurable?)

Who needs to be aware of childhood cancer?

Number of people in the specified age group alive at the end of each calendar year who had previously had a diagnosis of childhood cancer, Great Britain, 1961-2000



Insuring cancer survivors

- Risk of metachronous malignancy – predisposition syndromes
- Risk of second malignancy from therapy (radiotherapy, alkylators)
- Risk of problems from tumour and therapy – late effects
- Risks from new classes of molecular agents

Summary

- Rare but important cause of death
- Terminology in current usage now not relevant to diseases or their therapy
- Therapy will be expensive and will result in long term predisposition to disease
- Genetic component underestimated – impact for individual and family

Thank you for the invitation and questions

